

and the curve crosses the x-axis before 1.55 s.

B1

3 (a) (i) force is the rate of change of momentum

B1

(ii) Force from B on body A is equal in magnitude but opposite in direction to force on B from A (forces act on different bodies)

B1

Forces are of the same type

B1

(b) (i) maximum force = $(210 \times 10^{-9}) \times (138 \text{ to } 145) \times 9.81$
 $= 2.84 \times 10^{-4} \text{ to } 2.99 \times 10^{-4} \text{ N}$

C1

A1

(ii) Initial speed ~ 0

C1

Maximum speed = the area under the $a - t$ graph
 $= 1.20 \text{ to } 1.32 \text{ m s}^{-1}$

C1

A1

(iii) ground (and Earth) gain momentum

M1

In equal and opposite to the change for the flea, so momentum conserved B1

4 (a) (i) angle subtended at centre of circle B1